MUHAMMAD ARHAM KHAN

PYTHON BACKEND DEVELOPER | FULL STACK PYTHON DEVELOPER | AI APPLICATIONS ENGINEER

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Python Backend Developer with 3+ years of experience building REST APIs, multi-tenant systems and automation tools using Django, Django REST Framework, FastAPI and Flask. Currently building backend systems for HIMS and LMS/CMS platforms designed for 100+ hospitals at Camden Health System. Strong in PostgreSQL, Redis, Celery, RabbitMQ, Docker and WebSockets. Also built AI applications using the OpenAI API and NLP, and I am learning AWS and LLMs. Looking for backend or full stack Python roles.

AREA OF EXPERTISE

- Python Backend Development
- REST API Development & Integration
- WebSockets & Real-Time Apps
- Multi-Tenant Architecture
- Django & Django REST Framework
- PostgreSQL, MySQL & MongoDB
- Docker, NGINX & Linux
- OpenAI API & NLP (Basic)
- FastAPI & Flask
- Redis, Celery & RabbitMQ
- Web Scraping & Automation
- Authentication & RBAC

TECHNICAL SKILLS

Languages: Python (primary), JavaScript, TypeScript, SQL, HTML, CSS; also C++, Java, PHP

Backend Frameworks: Django, Django REST Framework (DRF), FastAPI, Flask, Node.js, NestJS

APIs & Real-Time: REST APIs, RESTful API design, third-party API integration, WebSockets, Socket.IO, WebRTC

Databases: PostgreSQL, MySQL, MongoDB, Redis, Supabase; database design, ORM, query optimization

Async & Queues: Celery, RabbitMQ, background jobs, task scheduling

DevOps & Tools: Docker, NGINX, Linux (Ubuntu), Git, GitHub, Heroku, Netlify, Azure

Automation & Scraping: Playwright, Selenium, web scraping, Chrome extensions, workflow automation

AI / ML: OpenAI API, NLP, text classification, sentiment analysis, conversational AI, prompt and token optimization

Frontend: Angular, Next.js, Nuxt.js, React (basic), jQuery

Currently Learning: AWS (S3, EC2, API Gateway), generative AI / LLMs

PROFESSIONAL EXPERIENCE

Associate Software Engineer, Camden Health System (Pvt) Ltd **March 2025 – Present**

- Build and maintain backend services and REST APIs in Python and Django for multi-tenant HIMS and LMS/CMS platforms designed for 100+ hospitals.
- Work on API development, third-party integration, and secure authentication and data exchange.
- Help with database design, management and query optimization using PostgreSQL.
- Take part in code reviews, debugging and performance optimization; fix issues in existing projects.
- Built a scalable, Zoho/Gmail-like email platform with Django, PostgreSQL and MongoDB, with multi-tenant architecture, individual and organization accounts, role-based access control, DKIM/SPF authentication and external mail server integration.
- Built a full-stack video calling platform with WebRTC, Socket.IO and React, with peer-to-peer calls, responsive UI and real-time user display.
- Completed and improved a Lovable-built app by adding a Supabase backend, fixing workflows and adding custom features.

Python Developer, GoldBarTech Pvt Ltd **July 2024 – March 2025**

- Developed an Amazon web scraper in Python (Playwright/Selenium) that collects product data automatically every 12 hours.
- Created an AI model for big data classification and an OpenAI-based system to identify and manage emails for large-scale email marketing.
- Designed an SMTP email tool for automated bulk email communication.
- Developed a Flask-based PMTA server management agent for DKIM/SPF generation, domain hosting, file management and server automation.

Intern, Eziline Software House **Jul 2023 – Sep 2023**

- Worked on full-stack development using Angular and Nuxt.js; helped build client websites.
- Worked with senior developers to improve backend systems and optimize database queries.
- Tested and debugged web applications.

Web Developer, Arctic Solution Ltd **Mar 2023 – Apr 2023**

- Led development of an automation system that increased operational efficiency by 15%.
- Built dynamic, responsive websites for small businesses.

Teacher, Skills Science College **Jul 2022 – Mar 2023**

- Taught computer science, web development and programming to high school students; managed the department's budget.
- Organized workshops and seminars on AI and machine learning.

Web Developer, Mak Traders **Jul 2021 – Nov 2021**

- Built and maintained websites for small businesses, including e-commerce integration and technical support.

KEY ACHIEVEMENTS

- Designed a custom Chrome extension that works with Helium 10 and scrapes Amazon X-Ray sheet data (Keywords, Total Sellers, Search Volume, Average Reviews, Revenue and more) in one click. Created separate versions for the USA and UK markets and pushed the data to Google Sheets.
- Built an automated Amazon web scraper that collects product details every 12 hours.
- Created an AI model for classifying big data, improving data analysis.
- Designed an SMTP email sending tool for automated bulk communication.
- Built a centralized frontend on a main server with backend operations on client servers: server selection, file management, PMTA management, DKIM/SPF generation, statistics, domain hosting and email account creation.
- Built a secure, scalable multi-tenant email service for individuals and organizations (like Zoho or Gmail) with organization and user management, role-based access control and external mail server integration.
- Created a video calling application that lets four people share video, messages and screen.
- Built an FBR/PRAL invoicing integration using a Flask bridge service.
- Led the development of dynamic websites for small businesses; managed large-scale university events.

PROJECTS

- Django-based conversational AI platform ("Sages") with human-like responses and voice interaction.
- Dockerized therapy platform where therapists and patients schedule sessions and manage therapy activities (full-stack lead).
- Medical store management system with inventory tracking, billing and role-based access control.
- Python CCTV recognition app that detects people in video feeds and retrieves their database profiles.
- Selenium scraper that collected location data of all McDonald's branches in the USA for business analysis.
- Video calling web app with real-time invitations and voice/video streaming.
- Fixed UI/UX bugs and improved responsiveness in a React project (Squid Mails); set up and debugged Supabase databases; deployed apps on Netlify, GitHub and Heroku.

EDUCATION

Bachelor of Science in Computer Science, Capital University of Science and Technology **2020 – 2024**

- Specialized in web development, AI and cybersecurity.
- Final year project: sentiment analysis using machine learning and NLP.
- Coursework: Data Structures, Algorithms, Artificial Intelligence, Database Systems, Software Engineering, Cybersecurity.

COURSES & CERTIFICATIONS

- Completed a Token Optimization course: learned how to write prompts that use fewer tokens, which lowers LLM cost and improves response speed.

AWARDS & ACHIEVEMENTS

- NaSCon 2022 Representative; Annual Function Manager (2022–2024); Event Manager at CUST; award-winning photographer.

INTERESTS

Web development, artificial intelligence, machine learning, UI/UX design, photography and traveling.

LANGUAGES

Proficient in English, Chinese and Urdu.